Class Meeting Handout #10

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In a general acid or base catalysis mechanism, the proton transfer is involved in the transition state. Can we determine information about the structure of this transition state? Yes, we can. We now begin exploring the true power of physical organic chemistry.[1] No more review, except when we need to review.

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Transition States in Catalysis

A catalyst must be part of the rate-determining transition state. Let us now consider transition states for general acid or base catalysis and how we can investigate them. Along the way, we will hopefully deepen our understanding of this subject.[2,3,4,5]

Discussion: The Transition State – 20 minutes

The textbook has a good presentation on the subject of catalysis. Let us use it. A copy is available in the lounge (I hope.)

1. We will examine Figure 9.13 in the textbook and comment on the mechanisms presented and the types of catalysis that result.

2. Consider a reaction coordinate for general acid catalysis and examine the Marcus analysis[6] presented in Figure 1 for two cases of general acid catalysis.

   (a) How would the activation energy change when a stronger general acid is used?

   (b) As we change the acid strength, how does the rate change w.r.t. acid strength in each case?

   (c) What is the extent of proton transfer in the transition state in each case?

   (d) Could we differentiate between these two cases using a log-log plot of rate vs. K_a of the acid?

   (e) What would be the maximum and minimum slopes that you would expect for such a plot?

3. Observe the Brønsted catalysis law for acid and for base as presented in equation 1 on the next page. How does it fit with your observations above?

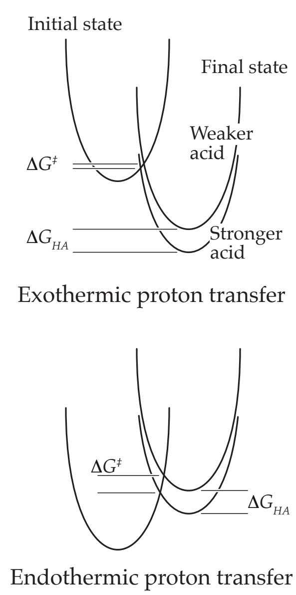

Figure 1: Marcus analysis of reaction coordinates for two cases of general acid catalysis.↑

[1] Read ch. 9.3 & 9.4 of the textbook.

[2] The concept of acid and base catalysis is not simple. The distinction between general and specific kinetics in water is often described as "subtle." Consider the following three contributions in which Eugene Kwan of Harvard presents some thoughts on teaching this idea, Addison Ault from the University of Toronto writes some critical but constructive comments, and then Kwan replies. All very respectful and thoughtful, the way science should be.

[3] "Factors Affecting the Relative Efficiency of General Acid Catalysis." E.E. Kwan, *J. Chem. Ed.*, 2005, *82*, 1026–1030. https://doi.org/10.1021/ed082p1026

[4] "General Acid and General Base Catalysis." A. Ault, *J. Chem. Ed.*, 2007, *84*, 38–39. https://doi.org/10.1021/ed084p38

[5] "The Rate of Proton Transfer." E.E. Kwan, *J. Chem. Ed.*, 2007, *84*, 39. https://doi.org/10.1021/ed084p39

[6] Marcus theory is explained in ch. 7.7 and applied to proton transfer in ch. 9.3.7

Exercise: A Brønsted Plot – 20 minutes

The Brønsted law is a log-log plot of the form…

$$\log k_{acid} = -\alpha \cdot pK_a + C.$$
$$\log k_{base} = \beta \cdot pK_a + C$$

Equation 1: The Brønsted equations for acid and base catalysis.
⟵
Remember that $pK_a = -\log K_a$ and so $-pK_a$ is actually just another way to say $\log K_a$. For the base equation, $K_a = K_W/K_b$ and so $pK_b = -\log K_a + \log K_W$. This is why there is no negative sign for β. See chapter 9.3.5 for more.

Let us consider the mutarotation of glucose, which is known to feature general acid and base catalysis.

4. Draw the mechanism for general acid and base catalyzed mutarotation of glucose. Identify the rate-determining step (rds.) Write the rate law.
5. Draw the structure of the transition state for the rds in each mechanism.
6. Construct a Brønsted plot using the data[7,8] from tables 1 and 2 on the following page.
7. Hypothesize the extent of proton transfer based on the slope of the Brønsted plot (α or β.)
8. Do you believe that a value of α or $\beta = 0.5$ means 50% proton transfer in the transitions state? How do you visualize 50%?
9. Draw a More O'Ferrall-Jencks plot and show what happens to the structure of the transition state as we change the strength of the acid or base. What is your observation? What does this mean for your interpretation of the Brønsted plots?

[7] "Contribution to the Theory of Acid and Basic Catalysis. The Mutarotation of Glucose."[8] J.N. Brönsted, E.A. Guggenheim, *J. Am. Chem. Soc.*, 1927, *49*, 2554–2584. https://doi.org/10.1021/ja01409a031.

[8] This work predates the field of physical organic chemistry. Louis Hammett reported his first efforts in acidity functions in 1928 and coined the name of the field in 1940. Perhaps Brønsted was the grandfather of it all? See "Louis Plack Hammett: 1894–1987" F.H. Westheimer, *Bioogr. Memoirs Natl. Acad. Sci.*, 1997, *72*, 137–150. https://doi.org/10.17226/5859

Learning Goals

There is more to a class meeting than what we do in 50 minutes of class time. After participating in the above exercises and completing the requirements described on the Moodle site, you will have achieved the following…

1. Understand the definitions of specific and general acid/base catalysis.
2. Understand the Brønsted relationship in acid/base catalysis and how Brønsted plots can be used to investigate acid/base-catalyzed reactions.

Looking Ahead

We have introduced the famous Brønsted plot, where the difference in free-energy transfer between a given acid and water ($K_a = e^{-\frac{\Delta G_{HA}}{RT}}$) is related to the free energy for proton transfer during the rate-determining step ($k = Ae^{-\frac{\Delta G^{\ddagger}}{RT}}$) of a reaction.

This was our first example of a linear free-energy relationship (LFER), which will be the focus of the next two weeks of this course. The idea of LFER is very important in physical organic chemistry. Everything that we discussed today will be used again and again.[9]

[9] Get ready for two weeks of LFER and other tools for determining reaction mechanism and transition state structure. We have built the foundation – now we will start building the house that Hammett built. Every class meeting is a brick in the wall.

Acid	K_a $/10^{-4}M$	k_{acid} $/10^{-3}M^{-1}min^{-1}$
Trimethylacetic acid	0.10	2.0
Propionic acid	0.14	2.1
Acetic acid	0.18	2.4
Phenylacetic acid	0.5	2.8
Glycolic acid	1.4	5.6
Formic acid	2.1	4.6
Mandelic acid	4.3	5.7
Chloroacetic acid	15	6.8

Table 1: General acid catalyzed mutarotation of glucose. Data from reference 7 ⟵

Conj. Acid of Base	K_a $/10^{-4}M$	k_{base} $/10^{-3}M^{-1}min^{-1}$
Trimethylacetic acid	0.10	31.4
Propionic acid	0.14	28.1
Acetic acid	0.18	26.5
Phenylacetic acid	0.5	20.0
Benzoic	0.6	15.2
o-Toluic acid	1.3	12.2
Glycolic acid	1.4	13.7
Formic acid	2.1	16.5
Mandelic acid	4.3	10.8
Salicylic acid	10	4.6
o-Chlorobenzoic acid	13	6.4
Chloroacetic acid	15	5.4
Cyanacetic acid	35	3.8

Table 2: General base catalyzed mutarotation of glucose. Data from reference 7 ⟵

The K_a of the conjugate acid of the base catalyst is presented in this table. No one uses K_b. A high value for K_a means that the basic form has a lower value for K_b. (The acid form is a stronger acid means that the basic form is a weaker base.) The is encapsulated by the statement that $K_a = {}^{K_W}/_{K_b}$.

Appendix: Python Code

Below is the *Python* code for expressing the Brønsted plot that we presented during this class meeting. Note that we are using the data from table 1 in the code example below. The interactive *Python* notebook will be available via a link[10] on the course Moodle page for this meeting.

[10] The notebook containing the code on this page can be accessed via Google *Colab* at https://colab.research.google.com/github/blinkletter/4410PythonNotebooks/blob/main/calcs/calcs_meeting_10.ipynb

```
from matplotlib import pyplot as plt
import numpy as np
from scipy.stats import linregress

data = [[0.10, 2.0],
        [0.14, 2.1],
        [0.18, 2.4],
        [0.5,  2.8],
        [1.4,  5.6],
        [2.1,  4.6],
        [4.3,  5.7],
        [15,   6.8]]

data = np.array(data)
K_a_array = data[:,0] * 1E-4              # Take the first item in each pair
k_acid_array = data[:,1] * 1E-3           # Take the second item in each pair

x = -np.log10(K_a_array)                  # pKa
y = np.log10(k_acid_array)                # log k_obs

# Here is the line fit. Just one command in python.
slope, intercept, r_value, p_value, std_err = linregress(x, y)

x_line = np.linspace(min(x), max(x), 100)  # Generate points for the fitted line
y_line = slope * x_line + intercept

plt.scatter(x, y)                         # Plot the data and the fitted line
plt.plot(x_line, y_line, color='red')

plt.xlabel('pKa')                         # Label the axes
plt.ylabel('log k')

plt.show()

print(f"slope = {slope:0.3f} +/- {std_err:0.3f}")
print(f"rsq = {r_value**2:0.4f}")
```

We will then use all the same code above but use the base-catalyzed data from table 2. Later in this course, you will see examples where we read data from a file. Mastering your tools will make your job easy.

```
data = [[0.10, 31.4],
        [0.14, 28.1],
        [0.18, 26.5],
        [0.5,  20.0],
        [0.6,  15.2],
        [1.3,  12.2],
        [1.4,  13.7],
        [2.1,  16.5],
        [4.3,  10.8],
        [10,   4.6 ],
        [13,   6.4 ],
        [15,   5.4 ],
        [35,   3.8 ]]
```